

Causality - SoSe 2025

Instrumental Variable Regression

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Throughout the lecture, we have touched the following topics:

- Observational Data vs. Randomized Controlled Trials
- Markov Properties and DAG Models (& d-Separation)
- Causal DAG Models and Causal Inference:
 - Parent-Adjustment
 - Path-Method
 - Back-Door and Front-Door Criterion
 - do-Calculus
- Causal Discovery:
 - Faithfulness & Causal Minimality
 - Constraint-Based CD (SGS, PC)
 - Score-Based CD (GES)
 - Restricted Models (LiNGAM, RESIT)

In today's lecture, we will link out of distribution generalization/extrapolation to causal estimation:

- Extrapolation in machine learning
- Distributionally robust optimization
- Instrumental variable regression
- Anchor regression

Extrapolation

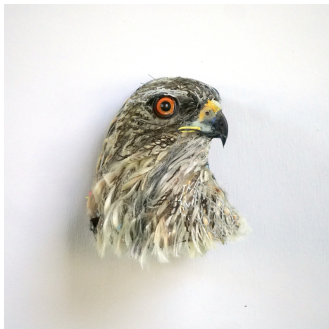


Figure 1: Sculpture by Tom Deininger

Observational Data



Figure 2: Sculpture by Tom Deininger

Observational data might show you exactly what you want to see, but when you try to **extrapolate** from it, things become tricky.

Extrapolation

We are interested in **extrapolation**!

That is, we aim to learn an estimator $\hat{\theta}$ that is **robust** to **some changes** in the test distribution, where $P_{\text{train}} \neq P_{\text{test}}$.



Figure 3: Tom Deininger

Distributional Robustness

Consider that we have a **random variable** $X \in \mathbb{R}^p$ and a **response** $Y \in \mathbb{R}$, where (X, Y) are drawn **iid** from some distribution P . Our task is to learn a prediction function $f_\theta : \mathcal{X} \rightarrow \mathcal{Y}$, parameterized by $\theta \in \mathbb{R}^q$.

Definition: In **empirical risk minimization** we aim to find θ^{ERM} , which is the minimizer of the **loss function** ℓ :

$$\operatorname{argmin}_{\theta} E_{(X,Y) \sim P}[\ell(Y, f_{\theta}(X))] .$$

Definition: A **distributionally robust** estimation has the target of inferring the θ with the **best worst-case** error for a **class of distributions** $\mathcal{Q}$ (which we will specify later)

$$\operatorname{argmin}_{\theta} \sup_{Q \in \mathcal{Q}} E_{(X,Y) \sim Q}[\ell(Y, f_{\theta}(X))] .$$

Real-World Applications

- **Autonomous driving:** As the world is a dynamic place, an autonomous car could not have seen every situation before: dirty or damaged signs on the road, unusual environments, . . .
- **Medicine:** Imagine that a classifier to detect cancer tissue is trained on data from a few big hospitals, but should be useful for other small hospitals with less data available, too.

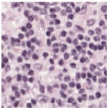
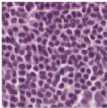
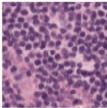
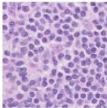
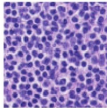
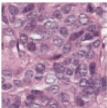
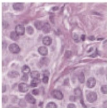
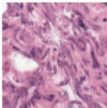
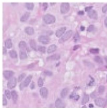
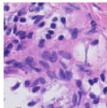
	Train			Val (OOD)	Test (OOD)
	d = Hospital 1	d = Hospital 2	d = Hospital 3	d = Hospital 4	d = Hospital 5
y = Normal					
y = Tumor					

Figure 4: Example from the WILDS dataset (Source: <https://wilds.stanford.edu>).

Example: Mean Shifts

When training a *regression model* via empirical risk minimization (ERM), we assume that:

$$P_{\text{train}} = P_{\text{test}}$$

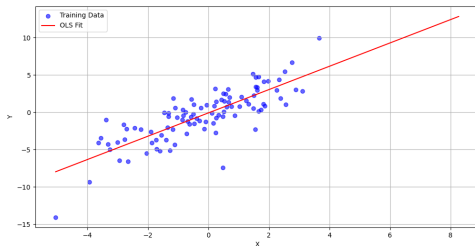


Figure 5: Shown is θ^{ERM} obtained from the training data P_{train} (in blue).

Example: Mean Shifts

But what happens if this assumption is violated? Given a class of distributions $\mathcal{Q}$, which includes both P_{train} and P_{test} , where $P_{\text{train}} \neq P_{\text{test}}$.

The data above was drawn from the following model¹ (which is **confounded due to H**):

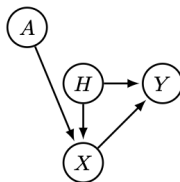
$$\epsilon_A, \epsilon_H, \epsilon_X, \epsilon_Y \sim \mathcal{N}(0, 1)$$

$$A := \epsilon_A$$

$$H := \epsilon_H$$

$$X := A + H + \epsilon_X$$

$$Y := X + 2H + \epsilon_Y$$



We performed OLS regression on $(X, Y) \sim P_{\text{train}}$ as defined above.

¹Rothenhäusler, et al. Anchor regression: heterogeneous data meet causality. J.R.Stat.Soc. 2021

Example: Mean Shifts

Consider now that the **test distribution** (in green) is affected by a **mean shift** in A induced by a **soft intervention**, i.e.

$$P_{\text{test}} = P_{\text{train}}^{\text{do}(A=\mathcal{N}(4,1))}.$$

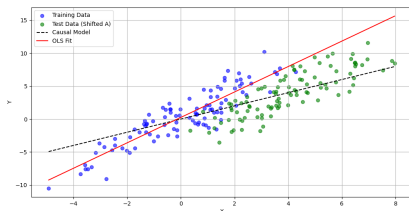
$$\epsilon_A, \epsilon_H, \epsilon_X, \epsilon_Y \sim \mathcal{N}(0, 1)$$

$$A := \epsilon_A + 4$$

$$H := \epsilon_H$$

$$X := A + H + \epsilon_X$$

$$Y := X + 2H + \epsilon_Y$$



- Clearly, the **OLS fit** is not good for the test data.
- We will see that a **causal model** trained, e.g., via instrumental variable regression **is more robust** to such shifts.

Invariance of Causal Model

Distributional Robustness: Overview

In the context of **regression**, our goal is to find θ^* so that

$$\theta^* = \operatorname{argmin}_{\theta} \sup_{Q \in \mathcal{Q}} E_{(X,Y) \sim Q} [(Y - f_{\theta}(X))^2] .$$

Related problems in machine learning are **domain adaptation**, **adversarial robustness**, **covariate shift**, etc. One prominent method to approach the problem is **distributionally robust optimization (DRO)**:²

$$\operatorname{argmin}_{\theta} \sup_{Q: D_{\text{KL}}(P_{\text{Train}} \| Q) \leq \delta} E_{(X,Y) \sim Q} [(Y - f_{\theta}(X))^2] .$$

Causal view: $\mathcal{Q}$ is constructed through **interventions** on the system.

²Duchi and Namkoong. *Learning models with uniform performance via distributionally robust optimization*. Ann. Stat., 2021

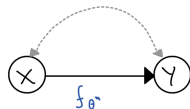
Structural Causal Model (1):

$$X := \epsilon_X$$

$$Y := f_{\theta^*}(X) + \epsilon_Y$$

where $E[\epsilon_Y] = 0$ and f_{θ^*} is the **causal function**.

Graph:



- We define $\mathcal{Q}_I := \{P^{do(X=x_0)} : x_0 \in \mathbb{R}\}$ as the set of distributions that can be constructed by **hard interventions on X**.
- Further note that the **causal function is invariant** to any interventions on X: $Y - f_{\theta^*}(X) = \epsilon_Y$.

Invariance of the Causal Function

Why are we interested in the **causal function**?

- Under the above assumptions, the **causal function** is **distributionally robust** with respect to any interventions on X !

Proposition: Under model (1), the causal function f_{θ^*} is distributionally robust under arbitrary interventions on X , i.e.,

$$f_{\theta^*} = \operatorname{argmin}_{\theta} \sup_{Q \in \mathcal{Q}_I} E_Q[(Y - f_{\theta}(X))^2] .$$

Note that we abbreviate $E_{(X,Y) \sim Q}$ with E_Q .

Due to the *invariance* of the causal function, note that for all $Q \in \mathcal{Q}_I$:

$$E_Q[(Y - f_{\theta^*}(X))^2] = E_Q[\epsilon_Y^2] = E[\epsilon_Y^2] .$$

Thus, it suffices to show that for any $\theta \in \Theta$:

$$\sup_{Q \in \mathcal{Q}_I} E_Q[(Y - f_{\theta}(X))^2] \geq E[\epsilon_Y^2] .$$

We have that:

$$\begin{aligned} E_Q[(Y - f_{\theta}(X))^2] &= E_Q[(f_{\theta^*}(X) + \epsilon_Y - f_{\theta}(X))^2] \\ &= E_Q[(f_{\theta^*}(X) - f_{\theta}(X))^2] + E[\epsilon_Y^2] + 2E_{Q_X}[(f_{\theta^*}(X) - f_{\theta}(X)) \cdot E_{Q_{Y|X}}[\epsilon_Y|X]] \\ &= E[\epsilon_Y^2] + E_Q[(f_{\theta^*}(X) - f_{\theta}(X) + E_{Q_{Y|X}}[\epsilon_Y|X])^2] - E_{Q_X}[E_{Q_{Y|X}}[\epsilon_Y|X]]^2 \\ &\geq E[\epsilon_Y^2] - E_{Q_X}[E_{Q_{Y|X}}[\epsilon_Y|X]]^2 \end{aligned}$$

Line two is due to the tower rule. Since any distribution $Q \in \mathcal{Q}_I$ induces an intervention on X , it removes any confounding between X and Y . Thus, for all such distributions $\epsilon_Y \perp\!\!\!\perp X$ implying $E_{Q_X}[E_{Q_{Y|X}}[\epsilon_Y|X]]^2 = E_Q[\epsilon_Y]^2 = 0$. Hence, it follows that

$$\sup_{Q \in \mathcal{Q}_I} E_Q[(Y - f_{\theta}(X))^2] \geq E[\epsilon_Y^2] + \sup_{Q \in \mathcal{Q}_I} (-E_{Q_X}[E_{Q_{Y|X}}[\epsilon_Y|X]]^2) .$$

Invariance of the Causal Function

We saw that the causal function has the lowest worst-case error for distributions that contain interventions on X .

Question: How do we learn it?

Question: Can't we just do empirical risk minimization?

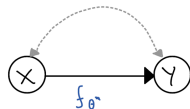
Confounding as a Source of Bias

Structural Causal Model (1):

$$X := \epsilon_X$$

$$Y := f_{\theta^*}(X) + \epsilon_Y$$

Graph:



where $E[\epsilon_Y] = 0$ and f_{θ^*} is the **causal function**.

Due to an **unobserved confounder** (no intervention on X), $E[\epsilon_Y|X] \neq 0$, which will induce a bias for ERM:

$$\begin{aligned} \operatorname{argmin}_{\theta} E[(Y - f_{\theta}(X))^2] &= E[Y|X] \\ &= E[f_{\theta^*}(X) + \epsilon_Y|X] \\ &= f_{\theta^*}(X) + \underbrace{E[\epsilon_Y|X]}_{\text{bias}} \end{aligned}$$

Linear IV Setting

Let's estimate the bias for **linear** models more explicitly!

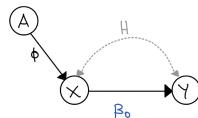
Structural Causal Model (2):

Graph:

$$A, H := \epsilon_A, \epsilon_H$$

$$X := \Phi A + \Gamma H + \epsilon_X$$

$$Y := \underbrace{\beta_0^T X}_{f_{\theta^*}(X)} + \underbrace{\eta^T H + \epsilon_Y}_{\epsilon'_Y}$$



All variables except Y can be multivariate, and we assume that $E[\epsilon_*] = 0$, and $E[\epsilon_Y|X] = 0$. However, note that $\epsilon'_Y \not\perp\!\!\!\perp X$!

Instrumental variable (IV) regression conditions:

- There is no path from A to Y that is not blocked by X ($A \perp\!\!\!\perp_{G_{\overline{X}}} Y$)
- $\text{Cov}(A, H) = 0$
- $\text{Cov}(A, X)$ has full column rank

$$\begin{aligned}\operatorname{argmin}_{\beta} E[(Y - \beta^T X)^2] &= (E[XX^T])^{-1} E[XY] \\&= (E[XX^T])^{-1} E[X(\beta_0^T X + \eta^T H + \epsilon_Y)] \\&= (E[XX^T])^{-1} (E[XX^T]\beta_0 + E[XH^T]\eta + \underbrace{E[\epsilon_Y]}_{=0}) \\&= \beta_0 + (E[XX^T])^{-1} E[(\Phi A + \Gamma H + \epsilon_X)H^T]\eta \\&= \beta_0 + \underbrace{(E[XX^T])^{-1} \Gamma E[HH^T]\eta}_{\text{bias}}\end{aligned}$$

Note that the term involving A disappears since by assumption $\operatorname{Cov}(A, H) = 0$, and $E[\epsilon_H] = 0$.

IV Regression

Angrist, Imbens, and Rubin (1995)

Instrumental Variable (IV) Regression

IV regression has an additional **constraint** which **ensures that the residuals** are **independent of A** , i.e.,

$$\text{Cov}(A, Y - \beta_{IV}^T X) = 0 ,$$

Intuitively, it exploits that the covariance between instrument A and confounder H is zero (Condition 2).

Instrumental Variable (IV) Regression

But does IV learn the true causal parameter?

$$\begin{aligned} \text{Cov}(A, Y - \beta_{IV}^T X) &= 0 \\ \Leftrightarrow \text{Cov}(A, Y) &= \beta_{IV}^T \text{Cov}(A, X) \\ \Leftrightarrow \text{Cov}(A, \beta_0^T X + \eta^T H + \epsilon_Y) &= \beta_{IV}^T \text{Cov}(A, X) \\ \Leftrightarrow \beta_0^T \text{Cov}(A, X) &= \beta_{IV}^T \text{Cov}(A, X) \end{aligned}$$

The last step is due to Condition 2. Further, since $\text{Cov}(A, X)$ has full column rank, we can conclude that $\beta_0^T = \beta_{IV}^T$.

Thus, **IV** learns the **causal** parameter.

Two-Stage Least Squares (2SLS)

Equivalently, we can exploit that conditioned on A , the expected error should be zero:

$$E[Y - \beta_{IV}^T X | A] = 0 .$$

Two-Stage Least Squares (2SLS) (Angrist, Imbens, 1995) is a practical method that exploits this property:

1. Regress X on A and obtain $\tilde{X} = E[X|A]$.
2. Regress Y on $\tilde{X}$ to obtain β_{IV} .

Note:

- Intuitively, the predictions $\tilde{X}$ are now uncorrelated to H .
- To not lose information about Y , it is important that $\text{Cov}(A, X)$ has full column rank (Condition 3).

Mean Shifts

Recall this example?

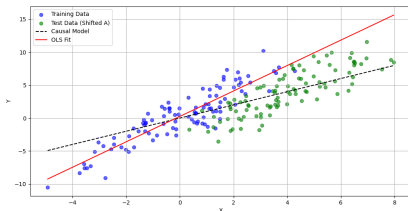
$$\epsilon_A, \epsilon_H, \epsilon_X, \epsilon_Y \sim \mathcal{N}(0, 1)$$

$$A := \epsilon_A + 4$$

$$H := \epsilon_H$$

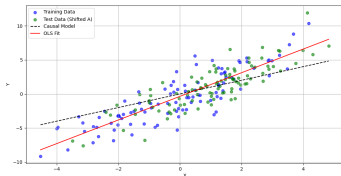
$$X := A + H + \epsilon_X$$

$$Y := X + 2H + \epsilon_Y$$

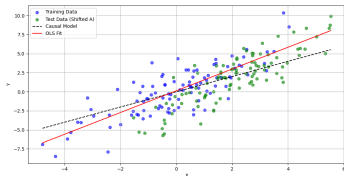


$\beta_{IV} = 1$ will be equal to the causal parameter (the dashed line)!

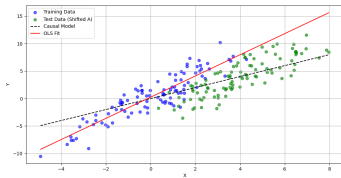
IV with Different Shifts



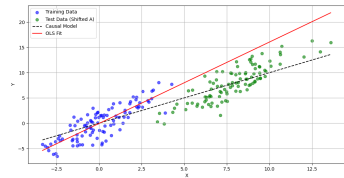
(a) Shift = 1



(a) Shift = 2



(a) Shift = 4



(a) Shift = 8

Limitations of IV

Despite the nice properties of IV that we discussed, IV also has limitations:

- IV Condition 3 requires $\#IVs \geq \#covariates$.
- All functional relations are **linear**.
- It only accounts for **mean shifts**.

IV and Distributional Robustness

In the context of distributional robustness (**extrapolation**), there are additional considerations with regard to learning the **causal function** f_{θ^*} :

- Identifying f_{θ^*} requires **strong assumptions** that are **hard to validate** for real-world datasets.
- f_{θ^*} may be **conservative for prediction under small shifts**.
- The invariance of f_{θ^*} does not hold for **interventions on Y or H** .

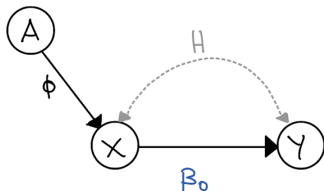
Some of these challenges are tackled by **anchor regression**!

Anchor Regression

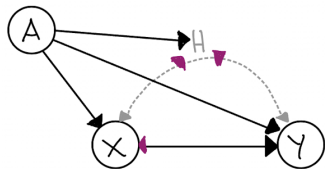
Rothenhäusler et al. (2021)

Anchor Regression

- The key idea of **anchor regression** is to find a **trade-off between OLS and IV**.
- It is robust to interventions on A of **strength** δ , i.e., $do(A = \delta)$.
- It only requires that A is a **source node** but drops the other assumptions of IV.



(a) IV graph



(a) Possible anchor graphs (subgraphs are allowed)

Anchor Regression

Formally, **anchor regression** minimizes the following objective:

$$\beta_{\text{anchor}}^{\gamma} = \operatorname{argmin}_{\beta} \underbrace{E[(Y - \beta^T X)^2]}_{\approx \text{OLS/ERM}} + (\gamma - 1) \underbrace{E[(E[Y - \beta^T X | A])^2]}_{\approx \text{IV}}$$

which resembles OLS regularized toward IV (encouraging uncorrelatedness of the residuals and A with larger γ).

Properties:

$$\begin{aligned}\beta_{\text{anchor}}^1 &= \beta_{\text{ERM}} = \operatorname{argmin}_{\beta} E[(Y - \beta^T X)^2] \\ \lim_{\gamma \rightarrow \infty} \beta_{\text{anchor}}^{\gamma} &= \beta_{\text{IV}} = \operatorname{argmin}_{\beta} E[(E[Y - \beta^T X | A])^2]\end{aligned}$$

where the latter only holds when the IV conditions are fulfilled.

It is possible to show that anchor regression **minimizes the worst-case loss** over interventions on A , i.e., $do(A = \delta)$. In particular,

$$\beta_{\text{anchor}}^\gamma = \operatorname{argmin}_\beta \sup_{\delta \in C^\gamma} E_{P^{do(A=\delta)}} [(Y - \beta^T X)^2]$$

where $C^\gamma = \{\delta : |\delta|_2 \leq \gamma\}$.

Guarantees on Distributional Robustness

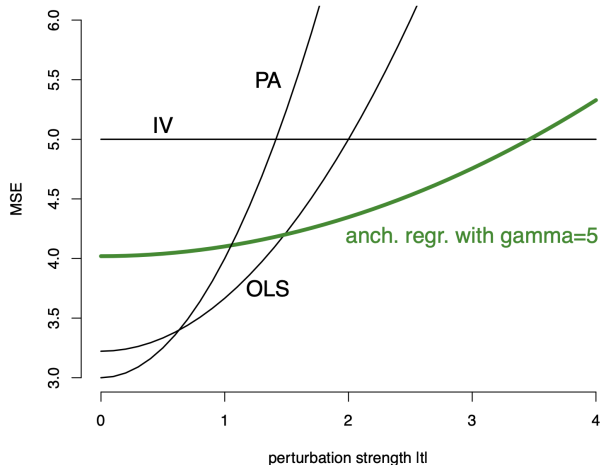


Figure 12: Test MSE under different perturbation strengths for $\gamma = 0$ (PA), IV, anchor regression with $\gamma = 5$, and OLS.

- **Recommended reading** is the script on instrumental variables by A.C. Cameron (<https://cameron.econ.ucdavis.edu/e240a/ch04iv.pdf>).
- Part of the material is building upon the lecture series “AI Center Projects in Machine Learning Research” (2022) at ETH Zurich.

Additional References

- Angrist, Imbens, Rubin. *Identification of Causal Effects Using Instrumental Variables*. J. Am. Stat. Assoc., 1995
- Angrist, Imbens. *Two-stage least squares estimation of average causal effects in models with variable treatment intensity*. J.Am.Stat.Assoc., 1995
- Rothenhäusler, Meinshausen, Bühlmann, Peters. *Anchor regression: Heterogeneous data meet causality*. J. R. Stat, 2021
- Bareinboim, Tian, Pearl. *Recovering from Selection Bias in Causal and Statistical Inference*. AAAI, 2014

Thank you!